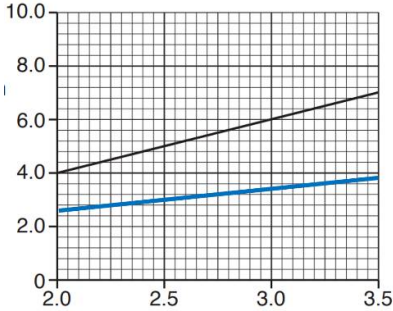


	Hence the work done required on the satellite is <u>positive</u> .	[–]
4(a)(i)	The waves reaching at AB will have <u>different path length</u>. When <u>path difference</u> is $n\lambda$, the waves will arrive at a point <u>in phase</u>, <u>superpose</u> and have <u>constructive</u> interference to form maxima. When path difference is $(n+0.5)\lambda$, the wave will arrive <u>anti-phase</u> at a point, <u>superpose</u> and have <u>destructive</u> interference to form minima.	[1] for path difference [1] for phase difference

4(a)(ii)	$I \propto A^2$ $\frac{I_1}{I_R} = \left(\frac{A_1}{A_R} \right)^2$ $\frac{I}{3I} = \left(\frac{1.5}{A_R} \right)^2 \Rightarrow A_R = 2.5981 \text{ cm}$ $A_Q = 2.5981 - 1.5 = 1.0981 \approx 1.1 \text{ cm}$	[1] for A_R [1] for answer
4(b)(i)	$ax = \lambda D \Rightarrow x = \frac{\lambda}{a} D$ Gradient of graph = $\frac{\lambda}{a}$ $\frac{(4.0 - 7.0) \times 10^{-3}}{(2.0 - 3.5)} = \frac{\lambda}{0.34 \times 10^{-3}}$ $\Rightarrow \lambda = 6.8 \times 10^{-7} \text{ m} = 680 \text{ nm}$	[1] for gradient relationship [1] for correct gradient [1] for answer
4(b)(ii)	 When a increases, gradient decreases, Lower intercept with vertical axis; smaller gradient	[1]
5(a)(i)	5.00	[1] ans